

APCyc: Property-Informed Design of Cyclic Peptides via Automated Cyclization

Yifan Zhao*

yzhao642@connect.hkust-gz.edu.cn

AI Thrust

The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China

Lang Qin*

lqin969@connect.hkust-gz.edu.cn

AI Thrust

The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China

Jintai Chen†

jintaichen@hkust-gz.edu.cn

AI-Peptide Drug Design Joint Laboratory
The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China

Abstract

Cyclic peptides represent a promising class of therapeutic compounds in modern drug discovery, often offering improved stability and binding affinity. However, the *de novo* design of cyclic peptides remains challenging because methods must identify pocket-adaptive cyclization patterns and linkage sites while simultaneously controlling drug-relevant properties. This challenge is particularly pronounced for recent generative models trained predominantly on linear peptide data, which may fail to capture cyclization-specific constraints. To address the limitation, we introduce **APCyc**, a target-aware *de novo* cyclic peptide generation framework that explicitly models cyclization and jointly optimizes multiple essential physicochemical properties. By using an expanded **residue vocabulary** and explicitly encoding **cyclization-site and linkage-type information**, APCyc learns cyclization-aware representations and leverages **Bayesian posterior guidance** to steer sampling toward cyclic peptides satisfying multiple property objectives. Experimental results demonstrate that our model learns target-dependent cyclization preferences, and enables effective and controllable multi-property optimization for cyclic peptide design. The source code of this paper is available at <https://github.com/HKUSTGZ-ML4Health-Lab/APCyc>.

CCS Concepts

• **Applied computing** → **Bioinformatics; Computational biology**; • **Computing methodologies** → *Artificial intelligence; Machine learning*.

Keywords

Cyclic Peptide; Generative Model; Diffusion Model

Accepted at the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026). DOI: 10.1145/3770855.3818908.

1 Introduction

Cyclic peptides are short and synthetically accessible amino acid sequences characterized by macrocyclic structural constraints that pre-organize the peptide backbone into bioactive conformations [16, 21], thereby minimizing the entropic cost of binding to target

proteins and making them attractive for therapeutic applications [14, 29, 36, 53, 58]. The advantages have motivated the development of computational methods for cyclic peptide discovery, with recent AI-based approaches increasingly used to model sequence-structure-property relationships and prioritize candidates from large combinatorial sequence spaces. However, designing therapeutically useful cyclic peptides remains challenging because candidate molecules must satisfy multiple coupled drug-relevant constraints beyond target binding, including solubility, permeability, protease resistance, metabolic stability, and safety-related properties [39]. Thus, cyclic peptide design is inherently a multi-objective optimization problem that requires methods capable of balancing therapeutic constraints rather than optimizing activity alone.

Recent target-aware peptide design methods have primarily focused on generating linear peptides conditioned on target (receptor) sequence and pocket geometry. Within this paradigm, geometry-aware generative models, including latent diffusion approach [27] or multimodal torsional flow matching [32, 35], have demonstrated the capability to perform full-atomic linear peptide design at target interfaces. Concurrently, approaches utilizing autoregressive generation [31], reinforcement learning paradigms [38], or pocket-conditioned diffusion models [46], have furthered the effective design of linear peptide binders across diverse targets [7]. However, these frameworks generally treat cyclization as a post-hoc structural modification rather than an integrated design variable [33, 43], failing to capture the complex, target-specific topological requirements inherent in high-affinity binding [43, 44, 54].

Beyond this lack of structural integration [22], existing frameworks rely on rigid, a priori cyclization heuristics [57]. In these methods, the cyclization type and linkage site positions are typically predefined based on heuristics, prior assumptions, or trial-and-error, rather than inferred directly from the structural requirements of the target binding pocket. This decoupling is fundamentally flawed, as binding sites exhibit substantial geometric diversity, ranging from deep and groove-like pockets to flat or shallow surfaces [17, 18] that inherently dictates the optimal peptide topology [24]. As a result, identifying the pocket-adaptive cyclization strategy in practical cyclic peptide discovery often requires costly enumeration of candidate cyclization strategies and iterative screening [25, 34], motivating the development of automated models for selecting cyclization strategies conditioned on the target binding site. Meanwhile, key drug-relevant properties such as permeability are tightly coupled to cyclization strategy, with the optimal choice varying across targets [8, 21]. In present methods, essential drug-like properties of cyclic peptides, including membrane permeability, solubility,

*Both authors contributed equally to this research.

†Corresponding author.



protease resistance, and immunogenicity, are largely overlooked, posing additional challenges for therapeutic peptide design.

To tackle the above challenges jointly, we propose **APCyc**, a generative framework that enables **A**utomated generation of cyclization linkage types and sites, as well as **P**roperty-informed optimization for **C**yclic peptide design. Unlike existing approaches that rely on fixed cyclization strategies [22, 57], APCyc explicitly formulates cyclization topology selection as a discrete, target-conditioned decision-making problem. This design enables the automated generation of cyclization types and residue-level linkage sites conditioned on the receptor pocket context. Specifically, APCyc decomposes cyclization topology into residue-level participation signals and pairwise linkage signals, allowing the model to explicitly learn cyclized residues, linkage sites, and linkage types during training. We first extend the peptide residue vocabulary by explicitly distinguishing residues involved in cyclization from non-cyclized residues, even when they share the same amino-acid identity. These cyclization-aware embeddings are then incorporated into the denoising network of a latent diffusion model [27], enabling supervised cyclization signals to condition and guide the denoising trajectory. For property-informed optimization, we train energy-based surrogates in latent space, which guide the diffusion process via gradients toward joint multi-property objectives. Figure 1 summarizes the high-level design concept of APCyc. We summarize our contributions as follows:

- **Automated pocket-conditioned cyclization** Unlike prior methods that require predefined cyclization types or linkage sites, APCyc learns to select both directly from the target binding-pocket context.
- **Property-informed controllable generation** APCyc uses Bayesian posterior guidance to steer cyclic peptide generation across affinity, permeability, protease resistance, solubility, and immunogenicity, achieving the best permeability proxy (0.107) and protease-resistance score (-1.474).
- **End-to-end cyclic peptide design** APCyc unifies cyclization selection, all-atom generation, and property-informed guidance. Across guidance settings, APCyc achieves leading structural quality, including the best Rosetta total score (-758.545) and consistency (0.971), while maintaining strong binding affinity.

2 Related Work

2.1 Target-aware cyclic peptide design

Traditional cyclic peptide discovery has relied largely on experimental screening methods, such as phage display technologies [19], and rational macrocyclization guided by medicinal-chemistry heuristics [12, 14, 58]. However, these approaches are labor-intensive and offer limited ability to jointly explore sequence, structure, and cyclization-topology spaces [37]. With the rise of deep learning and advances in protein structure prediction models such as AlphaFold2 [23], computational methods for cyclic peptide design have begun to emerge. AfCycDesign [43] adapts AlphaFold2 to cyclic peptides by encoding N-to-C terminal cyclization through modified positional encodings, enabling structure prediction, sequence redesign, and *de novo* hallucination of cyclic peptides. In parallel, diffusion-based generative models built on equivariant architectures [45]

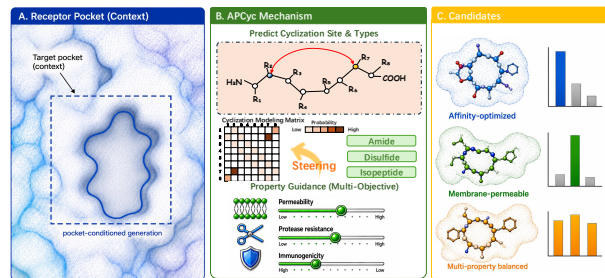


Figure 1: APCyc Concept. (A) The receptor pocket provides the structural context for target-conditioned cyclic peptide generation. (B) APCyc explicitly models cyclization as a discrete decision process by predicting both cyclization sites and linkage types. A cyclization modeling matrix encodes pairwise site probabilities, while chemically distinct linkage modes, including amide, disulfide, and isopeptide bonds, are incorporated into the generation process. Multi-objective property guidance further steers sampling toward desirable therapeutic profiles, such as membrane permeability, protease resistance, and low immunogenicity. (C) APCyc generates diverse candidate cyclic peptides under different design property objectives.

have improved geometric modeling for cyclic peptide generation. RFpeptides [44] further extends deep generative design to protein-binding macrocycles through a denoising diffusion-based pipeline. DiffPepBuilder [54] applies SE(3)-equivariant diffusion, with cyclization handled by post hoc disulfide bonding. Furthermore, CPSDE [57] utilizes chemical graph-based harmonic SDEs with atom-bond modeling, while CP-Composer [22] performs cyclic peptide design through geometric constraint composition. While effective, these methods are mainly trained on linear peptides and usually require cyclization constraints to be fixed a priori. In contrast, APCyc treats cyclization topology as a target-conditioned design variable, allowing it to infer cyclization types and linkage sites from the receptor pocket context.

2.2 Gradient-guided Bayesian diffusion

Diffusion models support controllable generation by modifying the reverse denoising process through guidance signals. Early work introduced classifier guidance [13], which steers sampling using gradients from a separately trained classifier that predicts the desired condition or attribute. Ho and Salimans [20] proposed classifier-free guidance, which enables conditional generation by combining conditional and unconditional score estimates without requiring an explicit classifier. Beyond categorical conditioning, guidance has been extended to energy-based formulations, where surrogate model gradients guide sampling toward desired objectives under Bayesian posterior interpretation [5, 10]. Further extensions have explored multi-constraint guidance, where multiple objectives or constraints are combined through score composition. For example, Bansal et al. [4] proposed universal guidance, which controls diffusion sampling using arbitrary guidance functions or off-the-shelf

auxiliary networks, while MolJO [41] applies gradient-based guidance to joint structure-based molecular optimization. Our work utilizes regressor-based gradient guidance, combining gradients from multiple physicochemical property surrogates as posterior guidance terms for joint optimization of drug-relevant properties.

3 Method

In this section, we present **APCyc**, a joint sequence–structure latent diffusion framework that enables automated cyclization-topology modeling and property-informed generation for cyclic peptides. Section 3.1 introduces the notation and formalizes target-conditioned cyclic peptide generation, then presents the joint sequence–structure latent diffusion formulation. Building on this formulation, Section 3.2 introduces **Automated Cyclization Pair Injection**, which injects cyclization-aware pair representations into the denoising network, enabling cyclization types and linkage sites to be inferred from the target context during generation. We then give the diffusion process in detail in Section 3.3. To promote desired drug-relevant property profiles, Section 3.4 presents **Bayesian Posterior Guidance**, which combines gradients from property surrogate models to steer the diffusion process toward joint multi-property objectives. Finally, Section 3.5 describes the training objectives and the complete inference algorithm. The overview of our method is shown in Figure 2.

3.1 Preliminaries

Geometric Graph Representation We represent the protein–peptide complex as a residue-level geometric graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ denotes the set of residue nodes from both the receptor and the peptide, and $\mathcal{E}$ denotes edges encoding sequential, spatial, and covalent relationships between residues. Each node $i \in \mathcal{N}$ is associated with a tuple $\mathbf{n}_i = (s_i, \mathbf{X}_i)$, where s_i denotes the residue token, $\mathbf{X}_i \in \mathbb{R}^{K \times 3}$ stores padded full-atom coordinates. Here, K is the maximum number of atoms per residue.

Following previous work [27, 35], we decompose the complex into a receptor graph $\mathcal{R}$ and a peptide graph $\mathcal{P}$. The receptor context $C \subseteq \mathcal{R}$ is defined as the binding-pocket residues within 10 Å of the reference peptide during training, and is treated as a fixed conditioning input during generation. We represent the cyclic peptide as $\mathcal{P} = (s, \mathbf{X})$, where s denotes the peptide sequence, $\mathbf{X}$ denotes the full-atom peptide structure.

In addition to sequence and structure, we denote the desired drug-relevant property targets as $\mathbf{Y} = (y_1, \dots, y_M) \in \mathbb{R}^M$, where each y_m is a normalized scalar score for a physicochemical or developability-related property, such as membrane permeability, solubility, or protease resistance. Given a fixed receptor context C and desired property targets $\mathbf{Y}$, conditional cyclic peptide generation aims to sample peptide graphs from $p(\mathcal{P} \mid C, \mathbf{Y})$, where sequence, structure, and cyclization linkage are generated jointly.

Latent Diffusion Modeling Following prior work on antibody and linear peptide design [26, 27], we adopt a dyMEAN-style autoencoder to map peptide graphs into residue-level latent representations. Specifically, an encoder $\mathcal{E}_\phi$ embeds the peptide graph into latent variables $\mathcal{Z} = \{(z_i, \tilde{z}_i)\}_{i=1}^L$, where z_i is E(3)-invariant and $\tilde{z}_i$ is E(3)-equivariant. We then learn a diffusion model over $\mathcal{Z}$ using

an equivariant GNN denoiser, and decode the final latents back to peptide sequence and structure via a decoder $\mathcal{D}_\xi$.

Cyclization-Enhanced Vocabulary Expansion Motivated by explicit cyclization annotations in cyclic peptide datasets such as CPSea [56], we extend the standard residue alphabet $\mathcal{A}_{\text{std}}$ to $\mathcal{A}_{\text{ext}} = \mathcal{A}_{\text{std}} \cup \mathcal{A}_{\text{cyc}}$, where $\mathcal{A}_{\text{cyc}}$ contains specialized tokens for residues participating in specific cyclization linkages, such as CYS_SS for disulfide-linked cysteine and LYS_ISO/ASP_ISO for residues involved in isopeptide bonds. This extension allows the model to distinguish residues involved in cyclization from their non-cyclized counterparts at the token level, even when they share the same standard amino-acid identity.

3.2 Automated Cyclization Pair Injection

At a discrete diffusion timestep $t \in \{0, \dots, T\}$, the noisy state $\mathcal{Z}_t = \{(z_{i,t}, \tilde{z}_{i,t})\}_{i=1}^L$ is sampled from the closed-form marginal of the DDPM-style latent diffusion process defined in Sec. 3.3:

$$z_{i,t} = \alpha_t z_{i,0} + \sigma_t \epsilon_i, \quad \tilde{z}_{i,t} = \alpha_t \tilde{z}_{i,0} + \sigma_t \tilde{\epsilon}_i, \quad (1)$$

where $\epsilon_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is Gaussian noise for the invariant latent, $\tilde{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is isotropic Gaussian noise with the same shape as $\tilde{z}_{i,0}$, and α_t and σ_t are the cumulative signal and noise coefficients satisfying $\sigma_t^2 = 1 - \alpha_t^2$.

Pair Representation Construction To facilitate cyclization inference, we construct a high-dimensional pairwise tensor $\mathbf{H}_t^{\text{pair}} \in \mathbb{R}^{L \times L \times d_p}$. We leverage both the invariant node latents $z_{i,t}$ and the equivariant coordinates $\tilde{z}_{i,t}$. The pairwise feature vector for residues (i, j) is computed as:

$$\mathbf{h}_{ij,t}^{\text{pair}} = \phi_{\text{enc}}(\text{Concat}(z_{i,t}, z_{j,t}, z_{i,t} \odot z_{j,t}, \psi(\|\tilde{z}_{i,t} - \tilde{z}_{j,t}\|))) \quad (2)$$

where $\odot$ denotes the Hadamard product and $\psi(\cdot)$ represents a sinusoidal RBF embedding [48]. This representation captures both the semantic compatibility (via z) and geometric proximity (via $\tilde{z}$) required for bond formation. To disentangle geometric modeling from cyclization reasoning, we project $\mathbf{h}_{ij,t}^{\text{pair}}$ into two task-specific subspaces:

$$\mathbf{h}_{ij,t}^{\text{struct}} = \phi_{\text{struct}}(\mathbf{h}_{ij,t}^{\text{pair}}), \quad \mathbf{h}_{ij,t}^{\text{cyc}} = \phi_{\text{cyc}}(\mathbf{h}_{ij,t}^{\text{pair}}). \quad (3)$$

where ϕ_{struct} and ϕ_{cyc} are separate projection heads (MLPs). The resulting $\mathbf{h}_{ij,t}^{\text{struct}}$ is utilized for geometric updates, while $\mathbf{h}_{ij,t}^{\text{cyc}}$ serves as the basis for predicting cyclization probability.

We then predict the cyclization type distribution $\mathbf{p}_{\tau,t}$ and the linkage matrix $\mathbf{S}_t$ at timestep t . Chemical constraints, such as Cys–Cys compatibility, are enforced through a validity mask $\mathbf{M}_{\text{valid}} \in \{0, 1\}^{L \times L}$. Distinct from row-wise attention, the linkage probabilities are computed by a global normalization over all valid residue-pair entries. Specifically, the cyclization head predicts pairwise site logits $\mathbf{L}_{\text{site},t} \in \mathbb{R}^{L \times L}$ from $\mathbf{h}_{ij,t}^{\text{cyc}}$. We first compute a masked, pre-symmetrized probability matrix $\mathbf{P}_t$ as

$$\text{vec}(\mathbf{P}_t) = \text{Softmax}(\text{vec}(\mathbf{L}_{\text{site},t} + \mathcal{M}(\mathbf{M}_{\text{valid}}))), \quad (4)$$

where $\text{vec}(\cdot)$ flattens an $L \times L$ matrix into a vector, and the mask operator $\mathcal{M}$ is defined element-wise as

$$\mathcal{M}(\mathbf{M}_{\text{valid}})_{ij} = \begin{cases} 0, & M_{\text{valid},ij} = 1, \\ -\infty, & M_{\text{valid},ij} = 0. \end{cases} \quad (5)$$

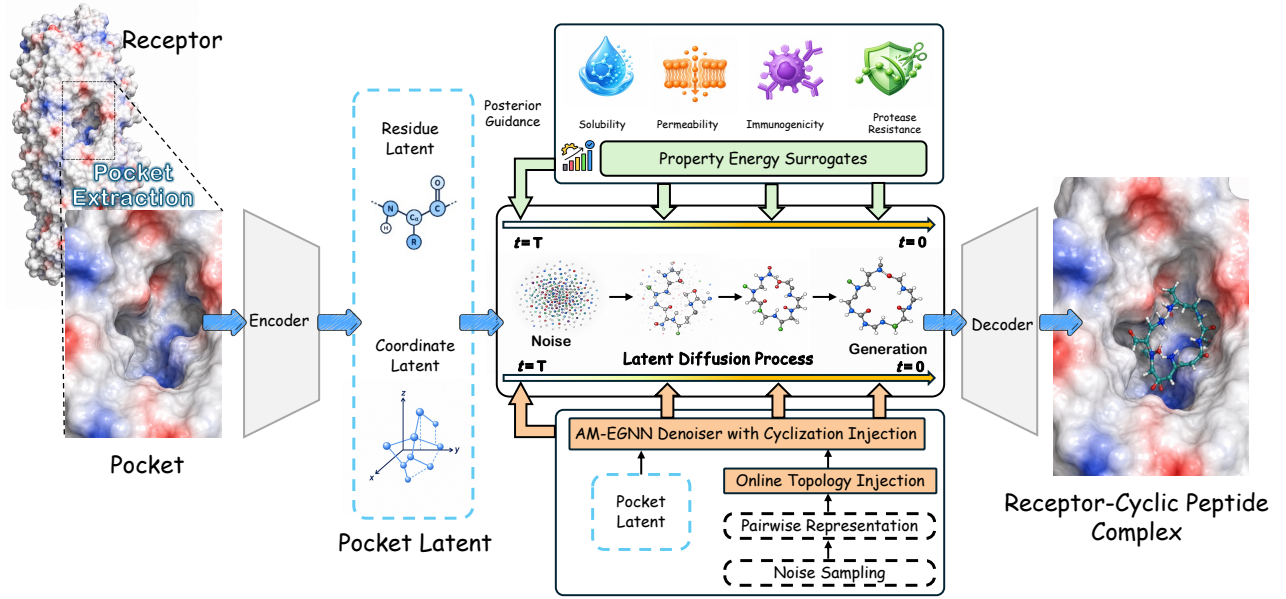


Figure 2: Sampling workflow of APCyc for pocket-conditioned cyclic peptide design. APCyc extracts a local receptor pocket from the input receptor structure and uses it as a fixed condition. Cyclic peptide variables, including residue identities, 3D coordinates, and cyclization topology, are encoded into residue and coordinate latents. A latent diffusion process with an AM-EGNN denoiser performs reverse sampling under the fixed pocket condition, with topology injection and energy guidance jointly steering cyclic peptide generation. The decoder reconstructs the final cyclic peptide ligand in the fixed receptor pocket.

Thus invalid entries receive zero probability after the softmax. The final undirected linkage matrix is obtained by symmetrizing $\mathbf{P}_t$:

$$S_{ij,t} = \begin{cases} P_{ij,t} + P_{ji,t}, & i \neq j, \\ 0, & i = j. \end{cases} \quad (6)$$

The cyclization type distribution is predicted as

$$\mathbf{p}_{\tau,t} = \text{Softmax} \left(\phi_{\text{type}} \left(\text{Pool} \left(\{\mathbf{h}_{ij,t}^{\text{cyc}}\}_{i < j} \right) \right) \right). \quad (7)$$

Adaptive Topology Injection via Edge Message Modulation

Given the predicted cyclization type distribution $\mathbf{p}_{\tau,t}$ and the symmetric linkage matrix S_t , we inject the online topology signal into the AM-EGNN denoising network through edge-feature augmentation and message modulation. The injection strength is controlled by timestep-dependent schedules $\lambda_{\text{feat}}(t)$, $\lambda_{\text{bias}}(t)$, and $\lambda_{\text{gate}}(t)$. In our implementation, these schedules are linear functions of the diffusion timestep and are used to gradually adjust the topology-injection strength during denoising.

Since S_t has already been symmetrized in Eq. (6), we directly use $S_{ij,t}$ as the undirected linkage probability between residues i and j . We first fuse the linkage probability and the cyclization type prediction into the edge features:

$$\mathbf{e}_{ij,t}^{\text{inj}} = \phi_{\text{edge}} \left(\text{Concat} \left(\mathbf{e}_{ij,t}^{\text{base}}, \mathbf{p}_{\tau,t}, \lambda_{\text{feat}}(t) S_{ij,t} \right) \right), \quad (8)$$

where $\mathbf{e}_{ij,t}^{\text{base}}$ denotes the base geometric RBF edge feature at timestep t , and $\lambda_{\text{feat}}(t) S_{ij,t}$ is treated as a scalar topology feature.

To further modulate message passing, we compute an additive bias and a multiplicative gate from the structural pair representation and the injected topology signal:

$$\mathbf{b}_{ij,t} = \phi_{\text{bias}} \left(\mathbf{h}_{ij,t}^{\text{struct}} \right) + \lambda_{\text{bias}}(t) S_{ij,t} \mathbf{1}_{d_m}, \quad (9)$$

$$\mathbf{g}_{ij,t} = \sigma \left(\phi_{\text{gate}} \left(\mathbf{h}_{ij,t}^{\text{struct}} \right) \right) \odot \left[\sigma \left(\lambda_{\text{gate}}(t) S_{ij,t} \right) \mathbf{1}_{d_m} \right], \quad (10)$$

where $\mathbf{1}_{d_m}$ is an all-one vector with the same dimension as the edge message, so that the scalar linkage probability is broadcast to the message dimension. Here $\sigma(\cdot)$ denotes the sigmoid function and $\odot$ denotes element-wise multiplication.

Within each AM-EGNN layer, let $\mathbf{m}_{ij,t}^{\text{base}}$ denote the base message from node j to node i computed using the injected edge feature $\mathbf{e}_{ij,t}^{\text{inj}}$. The topology-aware message is then defined as

$$\mathbf{m}_{ij,t}^{\text{mod}} = \left(\mathbf{m}_{ij,t}^{\text{base}} + \mathbf{b}_{ij,t} \right) \odot \mathbf{g}_{ij,t}. \quad (11)$$

This modulation allows the predicted cyclization topology to guide denoising while preserving the flexibility of the AM-EGNN to refine local geometry through the original geometric edge features.

PROPOSITION 3.1 (SE(3)-EQUIVARIANCE). Let F_θ denote the denoising update with topology injection. Then F_θ is SE(3)-equivariant, i.e.,

$$F_\theta(g \cdot \mathcal{Z}_t) = g \cdot F_\theta(\mathcal{Z}_t), \quad \forall g \in \text{SE}(3).$$

During the late denoising phase, we switch to a deterministic strategy by forming S_t^{inj} from the discretized linkage matrix S_t . This

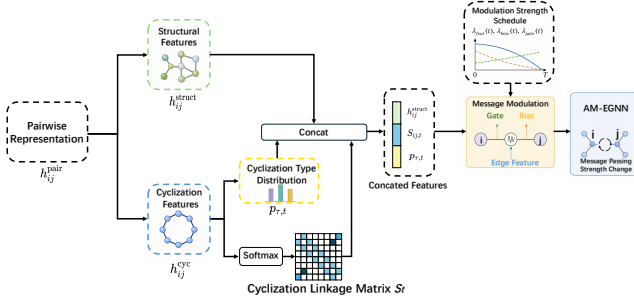


Figure 3: Cyclization topology information injection. APCyc predicts cyclization linkages and uses the topology signal to modulate AM-EGNN message passing.

"hard" linkage is treated as an explicit covalent edge, enhancing precise geometric closure in the final structure. Figure 3 illustrates the online topology injection module.

3.3 Geometry Latent Diffusion Process

Following Kong et al. [27], we model cyclic peptide generation with a DDPM-style latent diffusion process over residue-level latent variables. The encoder outputs $\mathbf{Z}_0 = \{(z_{i,0}, \tilde{z}_{i,0})\}_{i=1}^L$ are treated as clean latents, where $z_{i,0}$ is the invariant residue latent and $\tilde{z}_{i,0}$ is the equivariant geometric latent.

Let $t \in \{0, \dots, T\}$ be a discrete diffusion timestep. Given a variance-preserving noise schedule $\{\beta_t\}_{t=1}^T$, we define

$$\eta_t = 1 - \beta_t, \quad \bar{\eta}_t = \prod_{s=1}^t \eta_s, \quad \alpha_t = \sqrt{\bar{\eta}_t}, \quad \sigma_t = \sqrt{1 - \bar{\eta}_t}.$$

The forward perturbation admits the closed-form marginal

$$z_{i,t} = \alpha_t z_{i,0} + \sigma_t \epsilon_i, \quad \tilde{z}_{i,t} = \alpha_t \tilde{z}_{i,0} + \sigma_t \tilde{\epsilon}_i, \quad (12)$$

where ϵ_i and $\tilde{\epsilon}_i$ are independent standard Gaussian noise terms with shapes matching $z_{i,0}$ and $\tilde{z}_{i,0}$, respectively.

The denoising network $\epsilon_\theta(\mathbf{Z}_t, \mathbf{C}, t)$ predicts the noise added in Eq. (12). It is implemented as the cyclization-aware AM-EGNN described in Sec. 3.2. During generation, we start from $\mathbf{Z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and apply the standard DDPM reverse update

$$\mathbf{Z}_{t-1} = \frac{1}{\sqrt{\eta_t}} \left(\mathbf{Z}_t - \frac{\beta_t}{\sigma_t} \epsilon_\theta(\mathbf{Z}_t, \mathbf{C}, t) \right) + \sqrt{\beta_t} \xi, \quad \xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (13)$$

where $\tilde{\beta}_t = \frac{1-\bar{\eta}_{t-1}}{1-\bar{\eta}_t} \beta_t$, and ξ is set to zero when $t = 1$.

Training Objective At training time, we uniformly sample a timestep $t \in \{1, \dots, T\}$, construct $\mathbf{Z}_t$ using Eq. (12), and optimize the DDPM noise-prediction loss

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, \mathbf{Z}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{Z}_t, \mathbf{C}, t)\|_2^2], \quad (14)$$

where $\epsilon = \{(\epsilon_i, \tilde{\epsilon}_i)\}_{i=1}^L$.

We further supervise the online topology prediction using the pre-symmetrized linkage probability matrix $\mathbf{P}_t$ from Eq. (4) and the cyclization type distribution $\mathbf{p}_{r,t}$. Let $\mathbf{S}^*$ denote the ground-truth linkage matrix and τ^* denote the ground-truth cyclization type. We define

$$Y_{ij}^* = \frac{S_{ij}^* M_{\text{valid}, ij}}{\sum_{u,v} S_{uv}^* M_{\text{valid}, uv}},$$

and use

$$\mathcal{L}_{\text{topo}} = \mathbb{E}_{t, \mathbf{Z}_0, \epsilon} \left[- \sum_{i,j} Y_{ij}^* \log(P_{ij,t}) + \lambda_{\text{type}} \text{CE}(\mathbf{p}_{r,t}, \tau^*) \right]. \quad (15)$$

The final objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{topo}} \mathcal{L}_{\text{topo}}. \quad (16)$$

3.4 Property Optimization via Bayesian Posterior Guidance

To steer generation toward peptide candidates satisfying desired therapeutic criteria, such as binding affinity, permeability, solubility, or protease resistance, we perform property-guided sampling in the latent space. Given the target property vector $\mathbf{Y} = (y_1, \dots, y_M) \in \mathbb{R}^M$ and receptor context $\mathbf{C}$, the goal is to sample from the property-conditioned posterior $p_t(\mathbf{Z}_t | \mathbf{C}, \mathbf{Y})$ rather than only from the receptor-conditioned prior $p_t(\mathbf{Z}_t | \mathbf{C})$.

By Bayes' rule, the posterior score can be decomposed as

$$\nabla_{\mathbf{Z}_t} \log p_t(\mathbf{Z}_t | \mathbf{C}, \mathbf{Y}) = \nabla_{\mathbf{Z}_t} \log p_t(\mathbf{Z}_t | \mathbf{C}) + \nabla_{\mathbf{Z}_t} \log p_t(\mathbf{Y} | \mathbf{Z}_t, \mathbf{C}). \quad (17)$$

The first term is represented by the DDPM denoising network through $s_\theta(\mathbf{Z}_t, \mathbf{C}, t) \approx -\epsilon_\theta(\mathbf{Z}_t, \mathbf{C}, t)/\sigma_t$. The second term is intractable because drug-relevant properties are not directly defined on noisy latent states. We therefore introduce a time-conditioned differentiable surrogate Ψ_ψ and define the property energy

$$\mathcal{E}_\psi(\mathbf{Z}_t, t; \mathbf{Y}, \mathbf{C}) = \sum_{m=1}^M \lambda_m \ell_m(\hat{y}_{m,t}, y_m), \quad \hat{\mathbf{Y}}_t = \Psi_\psi(\mathbf{Z}_t, \mathbf{C}, t), \quad (18)$$

where $\hat{\mathbf{Y}}_t = (\hat{y}_{1,t}, \dots, \hat{y}_{M,t})$ is the predicted property vector at timestep t , ℓ_m measures the deviation from the m -th desired property target, and λ_m controls its relative importance. This energy serves as a differentiable proxy for the negative log-likelihood $-\log p_t(\mathbf{Y} | \mathbf{Z}_t, \mathbf{C})$. The resulting guided posterior is written as

$$p_t(\mathbf{Z}_t | \mathbf{C}, \mathbf{Y}) \propto p_t(\mathbf{Z}_t | \mathbf{C}) \exp(-\gamma_t \mathcal{E}_\psi(\mathbf{Z}_t, t; \mathbf{Y}, \mathbf{C})), \quad (19)$$

where γ_t is a timestep-dependent guidance scale. Therefore, the guided score is approximated by

$$s_{\theta, \psi}^{\text{guide}}(\mathbf{Z}_t, \mathbf{C}, \mathbf{Y}, t) = -\frac{\epsilon_\theta(\mathbf{Z}_t, \mathbf{C}, t)}{\sigma_t} - \gamma_t \nabla_{\mathbf{Z}_t} \mathcal{E}_\psi(\mathbf{Z}_t, t; \mathbf{Y}, \mathbf{C}). \quad (20)$$

Equivalently, under the DDPM noise-prediction parameterization, we use the guided noise estimate

$$\epsilon_{\theta, \psi}^{\text{guide}} = \epsilon_\theta + \sigma_t \gamma_t \nabla_{\mathbf{Z}_t} \mathcal{E}_\psi(\mathbf{Z}_t, t; \mathbf{Y}, \mathbf{C}). \quad (21)$$

During sampling, we replace ϵ_θ with $\epsilon_{\theta, \psi}^{\text{guide}}$ in the DDPM reverse update:

$$\mathbf{Z}_{t-1} = \frac{1}{\sqrt{\eta_t}} \left(\mathbf{Z}_t - \frac{\beta_t}{\sigma_t} \epsilon_{\theta, \psi}^{\text{guide}} \right) + \sqrt{\beta_t} \xi. \quad (22)$$

Thus, the energy gradient jointly guides the invariant residue latents $\{z_{i,t}\}_{i=1}^L$ and the equivariant geometric latents $\{\tilde{z}_{i,t}\}_{i=1}^L$, optimizing property with structural refinement during generation.

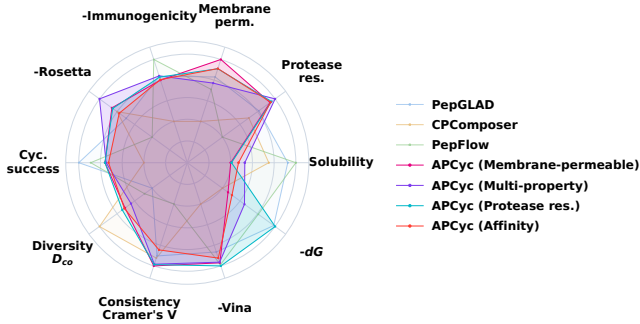


Figure 4: Overall comparison across different methods on therapeutic properties and generative metrics.

Algorithm 1 APCyc Inference with Topology and Property Guidance

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1: Input: Denoiser  $\epsilon_\theta$ , decoder  $\mathcal{D}_\xi$ , receptor context  $C$ , property targets  $Y = (y_1, \dots, y_M)$ , property surrogate  $\Psi_\psi$ , base weights  $w = \{w_m\}_{m=1}^M$ , validity mask  $M_{\text{valid}}$ , aggregation operator  $\text{Agg}(\cdot)$ , norm operator  $\text{Norm}(\cdot)$ , guidance scale  $\gamma_t$ , hard-linkage threshold  $t_{\text{hard}}$ 
2: Initialize: Sample  $Z_T = \{(z_{i,T}, \tilde{z}_{i,T})\}_{i=1}^L \sim \mathcal{N}(0, I)$ 
3: for  $t = T, \dots, 1$  do
4:    $p_{\tau,t}, P_t, S_t \leftarrow \text{CyclizationHeads}(Z_t, C, t; M_{\text{valid}})$ 
5:    $S_t^{\text{inj}} \leftarrow \begin{cases} \text{SymOneHot}(\arg \max_{i < j: M_{\text{valid},ij}=1} S_{ij,t}), & t \leq t_{\text{hard}}, \\ S_t, & t > t_{\text{hard}}. \end{cases}$ 
6:    $\hat{Y}_t \leftarrow \Psi_\psi(Z_t, C, t)$ 
7:   for  $m = 1, \dots, M$  do
8:      $\hat{E}_{m,t} \leftarrow \ell_m(\hat{y}_{m,t}, y_m)$ 
9:      $g_{m,t} \leftarrow \nabla_{Z_t} \hat{E}_{m,t}$ 
10:     $N_{m,t} \leftarrow \text{Norm}(g_{m,t})$ 
11:   end for
12:    $\lambda_m(t) \leftarrow \text{stopgrad} \left( w_m \cdot \frac{\text{Agg}(\{N_{j,t}\}_{j=1}^M)}{N_{m,t} + \epsilon_{\text{bal}}} \right), \quad m = 1, \dots, M$ 
13:    $\hat{E}_{\text{bal}}(Z_t, t) \leftarrow \sum_{m=1}^M \lambda_m(t) \hat{E}_{m,t}$ 
14:    $g_{\text{prop},t} \leftarrow \nabla_{Z_t} \hat{E}_{\text{bal}}(Z_t, t)$ 
15:    $\hat{\epsilon}_\theta \leftarrow \epsilon_\theta(Z_t, C, t; S_t^{\text{inj}}, p_{\tau,t})$ 
16:    $\hat{\epsilon}_{\theta,\psi}^{\text{guide}} \leftarrow \hat{\epsilon}_\theta + \gamma_t \sigma_t g_{\text{prop},t}$ 
17:    $Z_{t-1} \leftarrow \text{DDPMReverseStep}(Z_t, \hat{\epsilon}_{\theta,\psi}^{\text{guide}}, t)$ 
18: end for
19:  $p_{\tau,0}, P_0, S_0 \leftarrow \text{CyclizationHeads}(Z_0, C, 0; M_{\text{valid}})$ 
20:  $\hat{\mathcal{P}} \leftarrow \mathcal{D}_\xi(Z_0)$ 
21: return  $\hat{\mathcal{P}}, \arg \max p_{\tau,0}, S_0$ 

```

3.5 Training and Inference

APCyc is trained in a decoupled two-stage paradigm. We first train a joint latent autoencoder to learn residue-level invariant and equivariant latents, and then train a DDPM-style latent diffusion model over the frozen latent space.

Stage 1: Joint Latent Autoencoding We train the encoder $\mathcal{E}_\phi$ and decoder $\mathcal{D}_\xi$ to map a peptide graph $\mathcal{P}$ into clean residue-level latents $Z_0 = \{(z_{i,0}, \tilde{z}_{i,0})\}_{i=1}^L$ and reconstruct the full-atom peptide.

The autoencoding objective is

$$\mathcal{L}_{\text{AE}} = \frac{1}{L} \sum_{i=1}^L (\mathcal{L}_{\text{rec}}(i) + \mathcal{L}_{\text{latent}}(i)). \quad (23)$$

The reconstruction term is

$$\mathcal{L}_{\text{rec}}(i) = \text{CE}(s_i, \hat{s}_i) + \|X_i - \hat{X}_i\|_F^2 + \gamma_{\text{phys}} \mathcal{L}_{\text{phys}}(i), \quad (24)$$

where s_i and $\hat{s}_i$ denote the ground-truth and reconstructed residue types, X_i and $\hat{X}_i$ denote the ground-truth and reconstructed full-atom coordinates, and $\mathcal{L}_{\text{phys}}$ enforces valid local peptide geometry, such as bond lengths and bond angles. The latent regularization term $\mathcal{L}_{\text{latent}}$ regularizes the invariant latent $z_{i,0}$ and the equivariant geometric latent $\tilde{z}_{i,0}$; the detailed autoencoder objective is provided in Appendix B.

Stage 2: Latent Diffusion Training In the second stage, the autoencoder is frozen and the denoising network ϵ_θ is optimized using the joint objective $\mathcal{L}_{\text{total}}$ in Sec. 3.3. This stage uses the DDPM noise-prediction objective over noisy latents Z_t , together with auxiliary topology supervision for the Online Topology Injection module in Sec. 3.2. Thus, the denoiser learns both latent geometry denoising and timestep-dependent cyclization topology prediction.

Inference with Adaptive Balanced Guidance For property-guided generation, we use the target property vector $Y = (y_1, \dots, y_M)$ defined in Sec. 3.1. At each reverse diffusion step t , the property surrogate Ψ_ψ predicts

$$\hat{Y}_t = \Psi_\psi(Z_t, C, t), \quad \hat{Y}_t = (\hat{y}_{1,t}, \dots, \hat{y}_{M,t}). \quad (25)$$

For each property target, we define a property-wise energy

$$\hat{E}_{m,t} = \ell_m(\hat{y}_{m,t}, y_m), \quad m = 1, \dots, M, \quad (26)$$

where ℓ_m measures the deviation from the desired m -th property target.

To balance multiple objectives, we compute the gradient norm of each property-specific energy:

$$g_{m,t} = \nabla_{Z_t} \hat{E}_{m,t}, \quad N_{m,t} = \text{Norm}(g_{m,t}). \quad (27)$$

The adaptive weight for the m -th property is

$$\lambda_m(t) = \text{stopgrad} \left(w_m \cdot \frac{\text{Agg}(\{N_{j,t}\}_{j=1}^M)}{N_{m,t} + \epsilon_{\text{bal}}} \right), \quad (28)$$

where w_m is the base importance weight, $\text{Agg}(\cdot)$ is an aggregation operator over gradient norms, and ϵ_{bal} is a small constant for numerical stability. The balanced property energy and its guidance gradient are then

$$\hat{E}_{\text{bal}}(Z_t, t) = \sum_{m=1}^M \lambda_m(t) \hat{E}_{m,t}, \quad g_{\text{prop},t} = \nabla_{Z_t} \hat{E}_{\text{bal}}(Z_t, t). \quad (29)$$

Following Sec. 3.4, we form the guided noise estimate as

$$\epsilon_{\theta,\psi}^{\text{guide}} = \epsilon_\theta + \gamma_t \sigma_t g_{\text{prop},t}, \quad (30)$$

where γ_t is the timestep-dependent guidance scale. The guided noise is then used in the DDPM reverse update in Eq. (22). The complete inference procedure is summarized in Algorithm 1.

Table 1: Property metrics (mean over valid samples). Higher is better for Solubility, Protease resistance, and membrane permeability (Pproxy); lower is better for Immunogenicity. Top-1 and Top-2 are highlighted by bold and underline, respectively.

Method	Solubility $\uparrow$	Protease res. $\uparrow$	Membrane perm. $\uparrow$	Immunogenicity $\downarrow$
Baselines				
PepGLAD	<u>1.593</u>	-1.555	-0.010	-9.097
CPComposer	1.520	-1.604	-0.307	-7.576
PepFlow	1.624	-1.735	-0.093	-9.654
Ours				
Membrane-permeable	1.374	-1.499	0.107	-8.960
Multi-property (joint)	1.428	-1.474	-0.051	<u>-9.102</u>
Protease res.-optimized	1.377	<u>-1.492</u>	0.045	-9.062
Affinity-optimized	1.404	-1.496	<u>0.046</u>	-8.977

4 Experiment

4.1 Experimental Setup

Dataset In this work, we utilize **CPCore**, a high-quality refined subset of the **CPSea** dataset [56]. CPSea is a large-scale collection of 2.71 million cyclic peptide–protein complexes curated from the AlphaFold Database (AFDB). **CPCore** is constructed through intersectional filtering to ensure biophysical fidelity and essential properties. To be specific, the dataset consists of **71,867** unique complexes, a scale suitable for end-to-end training of generative models from scratch. It covers diverse and classical cyclization topologies, including **mainchain** amide bonds, **disulfide** bonds, and side-chain **isopeptide** (specifically Lys-Asp/Asn) linkages. Following Yang et al. [56], we partition the dataset based on FoldSeek structural clusters to prevent data leakage. Of the 30,819 clusters in CPCore, we randomly reserve 616 clusters ($\sim 2\%$) for validation and use the remainder for training. For evaluation, we employ the Large Non-Redundant (LNR) dataset. After filtering for sequence identity ($<40\%$ via MMseqs2) and structural suitability, the final test set comprises 56 unique targets. Concretely, after the cluster-based split, the CPCore training set contains 70,284 complexes and the validation set contains 1,583 complexes.

Property Collection To facilitate multi-objective optimization and ensure therapeutic viability, we developed a comprehensive property evaluation pipeline. This pipeline integrates a diverse suite of computational tools to characterize the physicochemical, pharmacological, and binding profiles of the cyclic peptides. The following provides a detailed overview of the specialized property prediction tools, and we provide implementation details for property collection in Appendix C.

- **Physicochemical Descriptors:** Hydrophobicity is assessed via the Grand Average of Hydropathy (GRAVY) using the Kyte-Doolittle scale [28]. Additional rational descriptors, including partition coefficient (logP) [55], topological polar surface area (TPSA) [15], and atom-normalized TPSA (rTPSA) [15] are calculated via RDKit.
- **Solubility:** Intrinsic solubility is predicted via the CamSol method [49].
- **Protease resistance:** Proteolytic stability is assessed via ProsperousPlus [30] to estimate the susceptibility of the cyclic backbone to enzymatic cleavage.

- **Immunogenicity:** Potential binding and presentation to MHC molecules are predicted using BigMHC [1].
- **Binding Affinity:** Interface binding energetics are characterized by AutoDock Vina score [51] and Rosetta interface energy (*rosetta_dG*) [2] using the interface_analyze protocol.

Baselines. We compare our method with several representative state-of-the-art peptide generative models. PepGLAD [27] performs target-aware full-atom peptide design using geometric latent diffusion with a variational autoencoder for joint sequence–structure generation. Building upon the PepGLAD architecture, CP-Composer [22] enables zero-shot cyclic peptide generation by decomposing cyclization into composable geometric constraints. Since CP-Composer focuses on the transferability from linear to cyclic peptides, we directly employ its original linear peptide generation weights and apply head-to-tail cyclization constraints for our comparisons. Additionally, we evaluate PepFlow [32], a multi-modal conditional flow-matching model that jointly models residue orientation, torsion angles, and sequence for full-atom co-design. To ensure a fair and rigorous comparison, PepGLAD and PepFlow were retrained from scratch on the CPCore dataset using identical data splits and training protocols, while CP-Composer was implemented in its intended zero-shot capacity.

Metrics To evaluate the efficacy of our method and the therapeutic potential of the designed cyclic peptides, we first assess their pharmacological viability and structural rationality. Following the property evaluation pipeline described previously, we calculate a suite of **drug-like properties**, including physicochemical descriptors (LogP, TPSA, GRAVY), intrinsic solubility, protease resistance and immunogenicity. We define **Affinity** using the AutoDock Vina score [51], which measures empirical docking fitness, and the Rosetta interface energy (*dG*) [2], which quantifies the binding free energy between the peptide and its target.

Furthermore, we evaluate the generative performance using success rates and ensemble-level distributional metrics. A generated sample is considered **Cyclization Success** if the distance between the cyclization residues falls within the range of $[3.0, 8.0]$ Å, while **Energy Success** is defined by a negative interface binding energy ($dG < 0$). In addition to these criteria, we adopt the Rosetta total score [9] as a measure of **Stability**, reflecting the physical plausibility and energetic favorability of the peptide’s three-dimensional

Table 2: Main results (mean over valid samples). Top-1 and Top-2 are highlighted by bold and underline, respectively.

Method	Stability	Success		Diversity	Consistency	Affinity	
	Rosetta ↓	Cyc. ↑	Energy ↑	D_{co} ↑	Cramér’s V ↑	Vina ↓	dG ↓
Baselines							
PepGLAD	-735.528	0.973	0.883	0.669	0.948	-4.544	<u>-12.140</u>
CPComposer	-743.422	0.809	0.700	0.898	0.953	-3.686	-10.716
PepFlow	-713.165	<u>0.943</u>	<u>0.711</u>	0.702	0.828	<u>-4.772</u>	-11.711
Ours							
Membrane-permeable	<u>-747.898</u>	0.907	0.671	0.787	0.971	-4.738	-10.872
Multi-property (joint)	-758.545	0.904	0.664	0.761	0.967	-4.726	-11.328
Protease res.-optimized	-747.176	0.907	0.689	<u>0.799</u>	<u>0.968</u>	-4.793	-12.178
Affinity-optimized	-741.560	0.898	0.648	0.790	0.934	-4.653	-10.981

binding conformation. To quantify **Diversity**, we utilize the co-diversity metric $D_{co} = \sqrt{D_{seq} \times D_{struct}}$ [27], where D_{seq} and D_{struct} denote the fractions of clusters over valid samples obtained via single-linkage hierarchical clustering based on sequence distance (1 – similarity, threshold 0.4) and C_α RMSD (threshold 4.0 Å), respectively. The correspondence between sequence and structure spaces is further evaluated by **Consistency**, measured via Cramér’s V [11], which captures the correlation between sequence and structural cluster assignments. The detailed metrics are in Appendix C.

4.2 Main Results

For each receptor in test set, we generate 10 candidate peptides. Samples that pass both the cyclization and energy criteria are regarded as **valid**. For each metric, we first compute the mean over valid samples within the same receptor, and then average across receptors to obtain the final reported performance.

Since our method enables demand-driven cyclic peptide design, we report several representative therapeutic design objectives, including affinity-optimized peptides, protease-resistant peptides, highly membrane-permeable peptides, and multi-property jointly optimized peptides. We provide the detailed guidance proportion in Appendix D. As shown in Table 1, our method allows explicit control over multiple key physicochemical objectives, including solubility, protease resistance, membrane permeability, and immunogenicity. Property-specific optimization successfully steers generated peptides toward the intended design targets. Figure 4 visualizes these normalized cross-metric trends, showing that AP-Cyc variants occupy complementary regions of the therapeutic-generative trade-off space while improving their intended guidance targets. For instance, the membrane-permeable setting achieves the highest permeability proxy among all methods while maintaining comparable solubility and immunogenicity, indicating that permeability enhancement does not substantially compromise other safety-related attributes. Meanwhile, the multi-property joint optimization yields the best protease resistance and competitive immunogenicity, demonstrating the capability of our framework to balance trade-offs across multiple therapeutic constraints rather than optimizing a single metric in isolation. Protease resistance-

Table 3: Ablation on protease resistance and immunogenicity.

Method	Protease res. ↑	Immunogenicity ↓
Multi-property (guided)	-1.474	-9.102
Protease res.-optimized	-1.492	-9.062
Base (no guidance)	-1.519	-9.011

and affinity-oriented variants further preserve favorable physicochemical profiles, suggesting that targeted optimization remains compatible with overall molecular feasibility.

Notably, we also observe a mild decrease in solubility under guided optimization. This trend likely arises from the intrinsic physicochemical coupling between membrane permeability, hydrophobicity, and structural compactness, where promoting permeability oriented features can partially counteract solubility. Such behavior is consistent with known trade-offs in peptide drug design and further highlights the necessity of controllable multi-objective optimization [42]. Together, these results verify that our framework enables controllable and therapeutically meaningful modulation of peptide properties beyond baseline generative behavior.

Beyond property-level improvements, Table 2 shows that the generated peptides also retain strong performance under standard generative evaluation criteria. Specifically, our guided variants achieve consistently strong **Stability**, outperforming the baselines in most cases, with *Affinity-optimized* being the only exception that is slightly weaker than CPComposer. In addition, the *Membrane-permeable* variant attains the highest **Consistency** score, indicating improved sequence–structure agreement under permeability-oriented guidance. Meanwhile, the *Protease res.-optimized* variant achieves the best binding affinity among our models, yielding the most favorable **Vina** score and interface free energy (dG). Finally, our method maintains competitive **Diversity** across guidance settings, demonstrating that improved stability and affinity do not come at the expense of ensemble-level variation. This suggests that enhanced therapeutic property control remains compatible with generative quality and binding effectiveness. Overall, the proposed framework achieves balanced improvements across both functional therapeutic objectives and conventional generative metrics, supporting its suitability for practical cyclic peptide design.

Table 4: Target-level case study on two representative targets. Baseline mean denotes the PepFlow/PepGLAD mean.

Target	Method	Vina ↓	Imm. ↓	Perm. ↑	Prot. ↑
3rc4	APCyc	-4.960	-12.615	2.408	-1.637
3rc4	Baseline mean	-1.149	-10.636	0.806	-1.639
4xal	APCyc	-6.820	-12.360	3.015	-1.256
4xal	Baseline mean	-4.023	-10.688	0.917	-1.768

4.3 Target-level Case Study

Beyond the aggregate benchmark results, we further conduct target-level case studies to examine whether APCyc can generate cyclic peptides with favorable binding and multi-property profiles on individual targets. Specifically, we select two representative targets, 3rc4 and 4xal, and compare APCyc-generated cyclic peptides with baseline references on the same targets.

As shown in Table 4, APCyc consistently achieves stronger binding scores and more favorable multi-property profiles than the PepFlow/PepGLAD mean on both targets. For Vina docking score and immunogenicity, where lower values are preferred, APCyc obtains substantially lower scores. For membrane permeability and protease resistance, where higher values are preferred, APCyc also achieves better or comparable results. In particular, APCyc improves membrane permeability from 0.806 to 2.408 on 3rc4, and from 0.917 to 3.015 on 4xal. These target-level results suggest that the overall benchmark improvements of APCyc are reflected not only in averaged metrics, but also in representative individual cases.

4.4 Ablation

To evaluate the contribution of the proposed guidance mechanism, we conduct an ablation study by comparing the full APCyc model with a variant where the guidance component is removed. We further analyze the impact of guidance on key therapeutic properties through targeted ablations. As shown in Table 3, guided optimization achieves competitive or improved protease resistance and immunogenicity compared with the unguided baseline, demonstrating controllable regulation of stability- and safety-related properties. Meanwhile, Table 5 shows that the membrane-permeable variant substantially increases the permeability proxy relative to the baseline, confirming the effectiveness of guidance in steering permeability-related behavior. Overall, these results highlight that explicit guidance is essential for directing cyclic peptide generation toward desired physicochemical objectives.

Table 5: Ablation on membrane permeability proxy. Higher values indicate better permeability.

Method	Permeability proxy ↑
Membrane-permeable (guided)	0.107
Multi-property (guided)	-0.051
Base (no guidance)	0.042

5 Limitations and Ethical Considerations

Data and supervision constraints APCyc is trained on CPCore [56], a curated CPSea subset derived from large-scale structure prediction resources. Although filtering and clustering improve data

quality, the complexes may still inherit prediction noise, imperfect pocket geometries, and target-dependent biases. In addition, CPCore mainly covers classical cyclization chemistries, including backbone amide, disulfide, and isopeptide linkages. Generalization to alternative crosslinkers, bicyclic topologies, or non-canonical stapling chemistries may therefore require additional structural data and broader chemical annotations.

Intended use, dual-use, and uncertainty APCyc is intended to support benign therapeutic discovery by proposing cyclic peptide candidates for expert review and experimental validation, rather than serving as a source of clinical recommendations. Because generative models for bioactive molecule design may be misused to search for harmful activity [52], practical deployment should remain within controlled research settings, follow biomedical governance and biosafety procedures, and include risk assessment, domain-expert oversight, and safety screening before synthesis [40, 47, 50]. In addition, safety-related objectives such as immunogenicity rely on computational predictors and may be uncertain or miscalibrated [6]. These predictions should therefore be treated as preliminary signals and complemented with additional models, expert assessment, experimental validation, and uncertainty reporting where possible [3].

6 Conclusions

We present APCyc, a target-aware latent diffusion framework that enables automated cyclization and property-informed design for cyclic peptides. APCyc dynamically infers pocket-adaptive linkage types and sites conditioned on the receptor context, and simultaneously optimizes multiple drug-like properties using the proposed Bayesian Posterior Guidance. As a result, APCyc addresses the critical challenge of controllable cyclic peptide generation under complex property constraints, providing a robust foundation and new insights for the design of next-generation therapeutic peptides.

Acknowledgments

This work was supported by the Guangdong Basic and Applied Basic Research Foundation (2026A1515011793), and the Youth S&T Talent Support Programme of Guangdong Provincial Association for Science and Technology (SKXRC2025467).

GenAI Disclosure

Parts of the manuscript preparation, including language refinement and editing, were assisted by generative AI tools. All scientific content, experimental design, implementation, and conclusions were developed and verified solely by the authors.

References

- [1] Benjamin Alexander Albert, Yunxiao Yang, Xiaoshan M Shao, Dipika Singh, Kellie N Smith, Valsamo Anagnostou, and Rachel Karchin. 2023. Deep neural networks predict class I major histocompatibility complex epitope presentation and transfer learn neoepitope immunogenicity. *Nature machine intelligence* 5, 8 (2023), 861–872.
- [2] Rebecca F Alford, Andrew Leaver-Fay, Jeliazko R Jeliazkov, Matthew J O’Meara, Frank P DiMaio, Hahnbeom Park, Maxim V Shapovalov, P Douglas Renfrew, Vikram K Mulligan, Kalli Kappel, et al. 2017. The Rosetta all-atom energy function for macromolecular modeling and design. *Journal of chemical theory and computation* 13, 6 (2017), 3031–3048.
- [3] Valentin Amrhein, Sander Greenland, and Blake McShane. 2019. Scientists rise up against statistical significance. *Nature* 567, 7748 (2019), 305–307.

- [4] Arpit Bansal, Hong-Min Chu, Avi Schwarzschild, Soumyadip Sengupta, Micah Goldblum, Jonas Geiping, and Tom Goldstein. 2023. Universal guidance for diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 843–852.
- [5] Fan Bao, Min Zhao, Zhongkai Hao, Peiyao Li, Chongxuan Li, and Jun Zhu. 2022. Equivariant energy-guided sde for inverse molecular design. *arXiv preprint arXiv:2209.15408* (2022).
- [6] Edmon Begoli, Tanmoy Bhattacharya, and Dimitri Kusnezov. 2019. The need for uncertainty quantification in machine-assisted medical decision making. *Nature Machine Intelligence* 1, 1 (2019), 20–23.
- [7] Suhaas Bhat, Kalyan Palepu, Lauren Hong, Joey Mao, Tianzheng Ye, Rema Iyer, Lin Zhao, Tianlai Chen, Sophia Vincoff, Rio Watson, et al. 2025. De novo design of peptide binders to conformationally diverse targets with contrastive language modeling. *Science Advances* 11, 4 (2025), eadr8638.
- [8] Arun Chandramohan, Hubert Josien, Tsz Ying Yuen, Ruchia Duggal, Diana Spiegelberg, Lin Yan, Yu-Chi Angela Juang, Lan Ge, Pietro G Aronica, Hung Yi Kristal Kaan, et al. 2024. Design-rules for stapled peptides with in vivo activity and their application to Mdm2/X antagonists. *Nature communications* 15, 1 (2024), 489.
- [9] Sidhartha Chaudhury, Sergey Lyskov, and Jeffrey J Gray. 2010. PyRosetta: a script-based interface for implementing molecular modeling algorithms using Rosetta. *Bioinformatics* 26, 5 (2010), 689–691.
- [10] Hyungjin Chung, Jeongsol Kim, Michael T Mccann, Marc L Klasky, and Jong Chul Ye. 2022. Diffusion posterior sampling for general noisy inverse problems. *arXiv preprint arXiv:2209.14687* (2022).
- [11] Harald Cramér. 1999. *Mathematical methods of statistics*. Vol. 9. Princeton university press.
- [12] Kaycie Deyle, Xu-Dong Kong, and Christian Heinis. 2017. Phage selection of cyclic peptides for application in research and drug development. *Accounts of chemical research* 50, 8 (2017), 1866–1874.
- [13] Prafulla Dhariwal and Alexander Nichol. 2021. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems* 34 (2021), 8780–8794.
- [14] Edward M Driggers, Stephen P Hale, Jinbo Lee, and Nicholas K Terrett. 2008. The exploration of macrocycles for drug discovery—an underexploited structural class. *Nature Reviews Drug Discovery* 7, 7 (2008), 608–624.
- [15] Peter Ertl, Bernhard Rohde, and Paul Selzer. 2000. Fast calculation of molecular polar surface area as a sum of fragment-based contributions and its application to the prediction of drug transport properties. *Journal of medicinal chemistry* 43, 20 (2000), 3714–3717.
- [16] Keld Fosgerau and Torsten Hoffmann. 2015. Peptide therapeutics: current status and future directions. *Drug discovery today* 20, 1 (2015), 122–128.
- [17] Diego Garcia Jimenez, Vasanathan Poongavanam, and Jan Kihlberg. 2023. Macrocycles in drug discovery—learning from the past for the future. *Journal of medicinal chemistry* 66, 8 (2023), 5377–5396.
- [18] Zuojun Guo, Bo Li, Li-Tien Cheng, Shenggao Zhou, J Andrew McCammon, and Jianwei Che. 2015. Identification of protein–ligand binding sites by the level-set variational implicit-solvent approach. *Journal of Chemical Theory and Computation* 11, 2 (2015), 753–765.
- [19] Christian Heinis, Trevor Rutherford, Stephan Freund, and Greg Winter. 2009. Phage-encoded combinatorial chemical libraries based on bicyclic peptides. *Nature chemical biology* 5, 7 (2009), 502–507.
- [20] Jonathan Ho and Tim Salimans. 2022. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598* (2022).
- [21] Xinjian Ji, Alexander L Nielsen, and Christian Heinis. 2024. Cyclic peptides for drug development. *Angewandte Chemie* 136, 3 (2024), e202308251.
- [22] Dapeng Jiang, Xiangzhe Kong, Jiaqi Han, Rui Jiao, Wenbing Huang, Stefano Ermon, Jianzhu Ma, and Yang Liu. 2025. Zero-Shot Cyclic Peptide Design via Composable Geometric Constraints. In *Proceedings of the 42nd International Conference on Machine Learning*.
- [23] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Židek, Anna Potapenko, et al. 2021. Highly accurate protein structure prediction with AlphaFold. *nature* 596, 7873 (2021), 583–589.
- [24] Kelly L Keeling, Okki Cho, Denis B Scanlon, Grant W Booker, Andrew D Abell, and Kate L Wegener. 2016. The key position: influence of staple location on constrained peptide conformation and binding. *Organic & biomolecular chemistry* 14, 41 (2016), 9731–9735.
- [25] Taegwan Kim, Eunbee Baek, and Jonghoon Kim. 2025. Exploring Macrocyclic Chemical Space: Strategies and Technologies for Drug Discovery. *Pharmaceuticals* 18, 5 (2025), 617.
- [26] Xiangzhe Kong, Wenbing Huang, and Yang Liu. 2023. End-to-end full-atom antibody design. *arXiv preprint arXiv:2302.00203* (2023).
- [27] Xiangzhe Kong, Yinjun Jia, Wenbing Huang, and Yang Liu. 2024. Full-atom peptide design with geometric latent diffusion. *Advances in Neural Information Processing Systems* 37 (2024), 74808–74839.
- [28] Jack Kyte and Russell F Doolittle. 1982. A simple method for displaying the hydrophobic character of a protein. *Journal of molecular biology* 157, 1 (1982), 105–132.
- [29] Jolene L Lau and Michael K Dunn. 2018. Therapeutic peptides: Historical perspectives, current development trends, and future directions. *Bioorganic & medicinal chemistry* 26, 10 (2018), 2700–2707.
- [30] Fuyi Li, Cong Wang, Xudong Guo, Tatsuya Akutsu, Geoffrey I Webb, Lachlan JM Coin, Lukasz Kurgan, and Jiangning Song. 2023. ProsperousPlus: a one-stop and comprehensive platform for accurate protease-specific substrate cleavage prediction and machine-learning model construction. *Briefings in Bioinformatics* 24, 6 (2023), bbad372.
- [31] Jiahua Li, Tong Chen, Shitong Luo, Chaoran Cheng, Jiaqi Guan, Ruihan Guo, Sheng Wang, Ge Liu, Jian Peng, and Jianzhu Ma. 2024. Hotspot-driven peptide design via multi-fragment autoregressive extension. *arXiv preprint arXiv:2411.18463* (2024).
- [32] Jiahua Li, Chaoran Cheng, Zuofan Wu, Ruihan Guo, Shitong Luo, Zhizhou Ren, Jian Peng, and Jianzhu Ma. 2024. Full-atom peptide design based on multi-modal flow matching. *arXiv preprint arXiv:2406.00735* (2024).
- [33] Jianan Li, Keisuke Yanagisawa, and Yutaka Akiyama. 2024. CycPeptMP: enhancing membrane permeability prediction of cyclic peptides with multi-level molecular features and data augmentation. *Briefings in Bioinformatics* 25, 5 (2024).
- [34] Xinting Li, Timothy W Craven, and Paul M Levine. 2022. Cyclic peptide screening methods for preclinical drug discovery: miniperspective. *Journal of Medicinal Chemistry* 65, 18 (2022), 11913–11926.
- [35] Haitao Lin, Odin Zhang, Huifeng Zhao, Dejun Jiang, Lirong Wu, Zicheng Liu, Yufei Huang, and Stan Z Li. 2024. Pflow: Target-aware peptide design with torsional flow matching. *arXiv preprint arXiv:2405.06642* (2024).
- [36] Lei Liu, Liu Yang, Suqi Cao, Zhigang Gao, Bin Yang, Guoqing Zhang, Ruixin Zhu, and Dingfeng Wu. 2024. CyclicPeptide: a knowledge base of natural and synthetic cyclic peptides. *Briefings in Bioinformatics* 25, 3 (2024), bbae190.
- [37] David J Newman and Gordon M Cragg. 2016. Natural products as sources of new drugs from 1981 to 2014. *Journal of natural products* 79, 3 (2016), 629–661.
- [38] Doogie Oh, Yongdoo Park, and Jongseong Kim. 2025. De novo generation of peptide binders with desired properties by deep generative models reinforced through enrichment of focused sets for iterative fine-tuning. *Chemical Communications* (2025).
- [39] Laszlo Otvos Jr and John D Wade. 2014. Current challenges in peptide-based drug discovery. 62 pages.
- [40] Jaspreet Pannu, Doni Bloomfield, Alex Zhu, Robert MacKnight, Gabe Gomes, Anita Cicero, and Thomas V Inglesby. 2024. Prioritizing high-consequence biological capabilities in evaluations of artificial intelligence models. *arXiv preprint arXiv:2407.13059* (2024).
- [41] Keyue Qiu, Yuxuan Song, Jie Yu, Hongbo Ma, Ziyao Cao, Zhilong Zhang, Yushuai Wu, Mingyue Zheng, Hao Zhou, and Wei-Ying Ma. 2024. Empower Structure-Based Molecule Optimization with Gradient Guided Bayesian Flow Networks. *arXiv preprint arXiv:2411.13280* (2024).
- [42] Theresa A Ramelot, Jonathan Palmer, Gaetano T Montelione, and Gaurav Bhardwaj. 2023. Cell-permeable chameleonic peptides: Exploiting conformational dynamics in de novo cyclic peptide design. *Current opinion in structural biology* 80 (2023), 102603.
- [43] Stephen A Rettie, Katelyn V Campbell, Asim K Bera, Alex Kang, Simon Kozlov, Yensi Flores Bueso, Joshmy De La Cruz, Maggie Ahlrichs, Suna Cheng, Stacey R Gerben, et al. 2025. Cyclic peptide structure prediction and design using AlphaFold2. *Nature Communications* 16, 1 (2025), 4730.
- [44] Stephen A Rettie, David Juergens, Victor Adebomi, Yensi Flores Bueso, Qinqin Zhao, Alexandria N Leveille, Andi Liu, Asim K Bera, Joana A Wilms, Alina Üffing, et al. 2025. Accurate de novo design of high-affinity protein-binding macrocycles using deep learning. *Nature Chemical Biology* (2025), 1–9.
- [45] Victor Garcia Satorras, Emiel Hoogeboom, and Max Welling. 2021. E(n) equivariant graph neural networks. In *International conference on machine learning*. PMLR, 9323–9332.
- [46] Tiara Natasha Binte Sayuti, Kakuly Mittal, Tan Lai Heng, and Jagath C Rajapakse. 2025. Target-aware latent diffusion model for design of apoptosis-inducing anticancer peptides. *Computers in Biology and Medicine* 199 (2025), 111286.
- [47] Petra Schneider, W Patrick Walters, Alleyn T Plowright, Norman Sieroka, Jennifer Listgarten, Robert A Goodnow Jr, Jasmin Fisher, Johanna M Jansen, José S Duca, Thomas S Rush, et al. 2020. Rethinking drug design in the artificial intelligence era. *Nature reviews drug discovery* 19, 5 (2020), 353–364.
- [48] Kristof T Schütt, Huziel E Sauceda, P-J Kindermans, Alexandre Tkatchenko, and K-R Müller. 2018. SchNet—a deep learning architecture for molecules and materials. *The Journal of chemical physics* 148, 24 (2018).
- [49] Pietro Sormani, Francesco A Aprile, and Michele Vendruscolo. 2015. The CamSol method of rational design of protein mutants with enhanced solubility. *Journal of molecular biology* 427, 2 (2015), 478–490.
- [50] Eric J Topol. 2019. High-performance medicine: the convergence of human and artificial intelligence. *Nature medicine* 25, 1 (2019), 44–56.

- [51] Oleg Trott and Arthur J Olson. 2010. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of computational chemistry* 31, 2 (2010), 455–461.
- [52] Fabio Urbina, Filipa Lentz, Cédric Invernizzi, and Sean Ekins. 2022. Dual use of artificial-intelligence-powered drug discovery. *Nature machine intelligence* 4, 3 (2022), 189–191.
- [53] Patrick Vlieghe, Vincent Lisowski, Jean Martinez, and Michel Khrestchatsky. 2010. Synthetic therapeutic peptides: science and market. *Drug discovery today* 15, 1-2 (2010), 40–56.
- [54] Fanhao Wang, Yuzhe Wang, Laiyi Feng, Changsheng Zhang, and Luhua Lai. 2024. Target-specific de novo peptide binder design with DiffPepBuilder. *Journal of Chemical Information and Modeling* 64, 24 (2024), 9135–9149.
- [55] Scott A Wildman and Gordon M Crippen. 1999. Prediction of physicochemical parameters by atomic contributions. *Journal of chemical information and computer sciences* 39, 5 (1999), 868–873.
- [56] Ziyi Yang, Hanyuan Xie, Yinjun Jia, Xiangzhe Kong, Jiqing Zheng, Ziting Zhang, Yang Liu, Lei Liu, and Yanyan Lan. 2025. CPSea: Large-scale cyclic peptide–protein complex dataset for machine learning in cyclic peptide design. In *Advances in Neural Information Processing Systems*. Datasets and Benchmarks Track.
- [57] Xiaoyu Zhou, Meng Li, Yifan Xiao, Jing Li, Dong Xue, Zhen Zheng, Jian Ma, and Quanquan Gu. 2025. Designing Cyclic Peptides via Harmonic Stochastic Differential Equations with Atom–Bond Modeling. In *Proceedings of the 42nd International Conference on Machine Learning*.
- [58] Alessandro Zorzi, Kaycie Deyle, and Christian Heinis. 2017. Cyclic peptide therapeutics: past, present and future. *Current opinion in chemical biology* 38 (2017), 24–29.

A Proof of Proposition 3.1

PROOF. Let $g \in \text{SE}(3)$ denote a rigid transformation acting on equivariant coordinates as $g(\mathbf{x}) = \mathbf{R}\mathbf{x} + \mathbf{t}$ with $\mathbf{R} \in \text{SO}(3)$. Invariant latent features remain unchanged.

Invariant topology prediction. The pair representation in (2) depends only on invariant node latents and Euclidean distances, which are preserved under rigid motions. Therefore the predicted linkage matrix $\hat{\mathbf{S}}_t$ obtained via the masked Softmax in (4) is invariant:

$$\hat{\mathbf{S}}_t(g \cdot \mathcal{Z}_t) = \hat{\mathbf{S}}_t(\mathcal{Z}_t).$$

Invariant Topology Injection. The injected edge features, bias, and gating terms defined in Eqs. (8)–(10) depend only on invariant distances, $\hat{\mathbf{S}}_t$, and scalar schedules. Hence all modulation coefficients remain invariant under $\text{SE}(3)$ transformations.

Equivariance of coordinate updates. The AM-EGNN coordinate update takes the standard form

$$\tilde{\mathbf{z}}'_{i,t} = \tilde{\mathbf{z}}_{i,t} + \sum_j \alpha_{ij,t} (\tilde{\mathbf{z}}_{i,t} - \tilde{\mathbf{z}}_{j,t}),$$

where the scalar weights $\alpha_{ij,t}$ depend only on invariant quantities. Applying g to both sides yields

$$\tilde{\mathbf{z}}'_{i,t}(g \cdot \mathcal{Z}_t) = g \cdot \tilde{\mathbf{z}}'_{i,t}(\mathcal{Z}_t),$$

which proves $\text{SE}(3)$ -equivariance.

Conclusion. Since topology prediction is invariant and coordinate updates are equivariant, the overall topology injection mechanism preserves $\text{SE}(3)$ -equivariance. $\square$

B Variational Autoencoder Training Loss

Following Kong et al. [27], we train a joint sequence–structure variational autoencoder that maps a peptide $\mathcal{P} = \{(s_i, \mathbf{X}_i)\}_{i=1}^L$ to residue-level latents $\mathcal{Z} = \{(z_i, \tilde{z}_i)\}_{i=1}^L$, where z_i is $\text{E}(3)$ -invariant and $\tilde{z}_i$ is $\text{E}(3)$ -equivariant. The decoder reconstructs $\hat{s}_i$ and $\hat{\mathbf{X}}_i$.

Overall objective. Following Sec. 3.5, the autoencoder is optimized by

$$\mathcal{L}_{\text{AE}} = \frac{1}{L} \sum_{i=1}^L (\mathcal{L}_{\text{rec}}(i) + \mathcal{L}_{\text{KL}}(i)). \quad (31)$$

Reconstruction loss. We use cross-entropy for residue tokens and MSE for full-atom coordinates, together with an auxiliary physical regularization:

$$\mathcal{L}_{\text{rec}}(i) = H(s_i, \hat{s}_i) + \|\mathbf{X}_i - \hat{\mathbf{X}}_i\|_2^2 + \gamma \mathcal{L}_{\text{phys}}(i). \quad (32)$$

Auxiliary physical loss. To better preserve realistic geometry, $\mathcal{L}_{\text{phys}}$ supervises C_α coordinates, bond lengths, and side-chain dihedral angles:

$$\mathcal{L}_{\text{phys}}(i) = \lambda_{\text{CA}} \mathcal{L}_{\text{CA}}(i) + \lambda_{\text{bond}} \mathcal{L}_{\text{bond}}(i) + \lambda_{\text{angle}} \mathcal{L}_{\text{angle}}(i). \quad (33)$$

Latent regularization. The invariant latent z_i is regularized toward a standard Gaussian, while the equivariant latent $\tilde{z}_i$ is anchored to the reference C_α coordinate $\mathbf{r}_i$:

$$\mathcal{L}_{\text{KL}}(i) = D_{\text{KL}}(q_\phi(z_i) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) + \eta \|\tilde{z}_i - \mathbf{r}_i\|_2^2. \quad (34)$$

Default weights. We set $\lambda_{\text{CA}} = 1.0$, $\lambda_{\text{bond}} = 1.0$, and $\lambda_{\text{angle}} = 0.5$ in experiments, and use a small η to stabilize equivariant latent learning.

C Metrics Implementation

C.1 Inputs and identifiers

Each candidate is provided as a complex PDB (optionally gzipped). We assume a two-chain convention in the complex: the peptide is chain L and the receptor/target protein is chain R. We use the PDB filename stem as the peptide identifier id. All sequence-based predictors operate on the peptide-chain sequence extracted from chain L; structure-based descriptors operate on the chain-L structure; affinity evaluates the L–R interface.

C.2 Solubility (CamSol intrinsic)

To assess solubility, we employ the CamSol-intrinsic predictor on the peptide sequence and record the returned scalar `camsol_score` (higher indicates better solubility):

$$S_{\text{sol}} := \text{camsol_score}. \quad (35)$$

C.3 Enzymatic stability / proteolysis susceptibility (ProsperousPlus)

To quantify proteolysis susceptibility, we employ ProsperousPlus in prediction mode to estimate cleavage propensity under a fixed panel of proteases

$$\mathcal{P} = \{\text{S01.001}, \text{A01.001}, \text{C01.060}, \text{M10.003}, \text{M10.008}\}. \quad (36)$$

Because ProsperousPlus expects a fixed-length context, we convert each peptide sequence into a set of 8-mer windows. Specifically, after uppercasing and replacing non-canonical characters with `-`, we generate length-8 windows with stride 2, ensuring the C-terminal window is included; if the number of windows exceeds 30, we subsample evenly to at most 30 windows. For each window set, ProsperousPlus outputs per-position cleavage probabilities (column

pro) for each protease. We aggregate these raw probabilities by averaging within each protease and then summing across proteases:

$$\text{proteolysis_sumscore} := \sum_{p \in \mathcal{P}} \left(\frac{1}{N_p} \sum_{k=1}^{N_p} \text{pro}_{p,k} \right), \quad (37)$$

where N_p is the number of probability entries produced for protease p across all windows/positions. Smaller values imply lower cleavage propensity (i.e., higher stability), and we therefore define a higher-is-better stability score as

$$S_{\text{prot}} := -\text{proteolysis_sumscore}. \quad (38)$$

C.4 Immunogenicity proxy (BigMHC EL/IM)

To assess immunogenicity, we run BigMHC on a specified MHC-I allele (default: HLA-A*02:01) and obtain two outputs per peptide: **e1** (presentation likelihood) and **im** (immunogenicity likelihood). Peptides of length 8–11 are scored directly. For peptides longer than 11 residues, we enumerate sliding windows of lengths 8, 9, 10, 11, score each window, and average the top- K window scores (we use $K=3$) for EL and IM separately. We report the raw immunogenicity log-score, where lower values indicate lower predicted immunogenicity, and use its sign-flipped form for internal surrogate training and guidance:

$$I_{\text{imm}} := \log(\max(\text{e1}, \epsilon)) + \log(\max(\text{im}, \epsilon)), \quad (39)$$

$$S_{\text{imm}} := -I_{\text{imm}}. \quad (40)$$

Thus all guidance objectives follow a higher-is-better convention.

C.5 Physicochemical descriptors from peptide-chain structure

To obtain structure-derived physicochemical descriptors, we analyze the peptide chain L extracted from the complex PDB. We compute: (i) **GRAVY** as the average Kyte–Doolittle hydrophathy over the extracted peptide sequence; (ii) **TPSA** and **logP** using RDKit descriptors computed from the peptide-chain PDB block; (iii) the heavy-atom-normalized polar surface area

$$\text{rTPSA} := \frac{\text{TPSA}}{\text{\#heavy atoms}}. \quad (41)$$

C.6 Permeability proxy score

We consolidate the above descriptors into a single permeability proxy. Since the raw descriptors have different scales, we first normalize each descriptor by dataset-level z-scoring:

$$z(x) = \frac{x - \mu(x)}{\sigma(x)}, \quad (42)$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ are computed over the dataset (ignoring missing values; using population standard deviation). We then define

$$P_{\text{proxy}} := z(\log P) + z(\text{GRAVY}) - z(\text{TPSA}) - z(\text{rTPSA}), \quad (43)$$

which corresponds to higher hydrophobicity and lower polar surface area being more favorable.

C.7 Binding affinity: Vina score-only evaluation

Binding affinity is estimated by evaluating the *given* complex pose without redocking. Chains L and R are extracted, converted to PDBQT via MGLTools. AutoDock Vina is executed in `-score_only` mode:

```
vina -receptor R.pdbqt -ligand L.pdbqt -center_* c_*
      -size_* s_* -score_only
```

The scoring box is defined from ligand bounds with padding p (default 10 Å):

$$c_x = \frac{x_{\min} + x_{\max}}{2}, \quad s_x = \max((x_{\max} - x_{\min}) + p, 1.0), \quad (44)$$

and analogously for y, z . The reported affinity (kcal/mol) is recorded as `vina_score`.

C.8 Binding affinity: RosettaScripts interface

In addition to Vina, we compute a Rosetta interface energy proxy with RosettaScripts. Unlike the example FastRelax workflow, we do not perform additional relaxation in this pipeline; instead, we directly evaluate the input complex using the `ref2015` scorefunction and a minimal protocol (`interface_analyze.xml`) that applies a single Ddg filter with `jump=1`, `repeats=1`, `repack=0` (no repacking/minimization), and scorefunction `ref2015`. We run `rosetta_scripts.*` in `score-only` mode and extract the scalar `dg` reported by the filter as `rosetta_dG`.

C.9 Affinity normalization and combined score

Because Vina and Rosetta energies have different scales, we convert each to a higher-is-better normalized term via z-scoring and sign flip. We then define

$$A_{\text{ros}} := -z(\text{rosetta_dG}), \quad A_{\text{vina}} := -z(\text{vina_score}), \quad (45)$$

$$A_{\text{aff}} := \frac{1}{2}A_{\text{ros}} + \frac{1}{2}A_{\text{vina}}, \quad (46)$$

and use A_{aff} as affinity property to train the surrogate.

C.10 Unified optimization direction

For surrogate training and gradient-based guidance, all property scores are converted to a *higher-is-better* convention to ensure a consistent optimization direction across objectives. This transformation is used only for internal optimization. Reported results follow the standard biochemical interpretation of each metric.

D Guidance Weights for Therapeutic Objectives

APCyc uses five Bayesian posterior-guided sampling configurations, **Base**, **Aff.**, **Prot.**, **Perm.**, and **Multi**, by assigning property-specific guidance weights. Table 6 lists the coefficients applied to the joint energy function in Sec. 3.4.

Table 6: Guidance weights for different therapeutic design objectives.

Obj.	Aff.	Prot.	Perm.	Sol.	Imm.
Base	0	0	0	0	0
Aff.	10	0	0	0	0
Prot.	8	8	0	2	0
Perm.	8	2	10	2	0
Multi	6	4	4	3	5